NATURAL LANGUAGE PROCESSING

FINAL EXAMINATION

Search Engine (100 Points)

Students are grouped into teams of five.

Goal

Develop a search engine, like Google Search, that incorporates the following NLP concepts:

- N-grams (for phrase matching, query expansion & features)
- Classification (query or document classification / filtering)
- TF-IDF (document representation and ranking)

Teams & Roles

- **Project Lead** – Coordinates and track progress of members, and manages final integration.
- **Pre-processing Engineer** – Scraping, Tokenization, n-grams, Stop words
- **Representation Engineer** – TF-IDF implementation & vector store
- **Classifier** – Builds and evaluates classification component
- **Frontend / API & Testing** - Search UI / API, Test Functionality

Required Deliverables

1. Code Repository (git)
    a. New repository with clear README and instructions to run;
    b. GitHub Project with tickets and assignments.

2. Training scripts
    a. Organize datasets for training and testing splits;
    b. Document and store trained models (could be linked to Google Drive folder if model exceeds 50mb).

3. Classification / Training
    a. Test & Provide Accuracy Report of the model

4. Search Application / API Server
    a. Must document required engine / platform and dependencies to run the application;

5. Presentation

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a. Importance of the project (real-world relevance)
b. Scope and limitations
c. System Architecture
    i. Diagram of the pipeline
    ii. Mapping and assignment of each module
    iii. Technologies used
d. Pre-processing & Indexing
e. Representation & Retrieval (TF-IDF)
f. N-grams (Autosuggest)
g. Classification
h. Search Interface Demo
i. Future Work

Grading Criteria:

- Group (70)

    - ✓ Indexing & Pre-processing (15)
    - ✓ TF-IDF Representation & Retrieval – Accuracy and Efficiency (20)
    - ✓ Classification – Accuracy (15)
    - ✓ Search Interface (20)
    - ✓ Bonus (+10) - Spell checking, containerized deployment, or advanced ranking methods

- Individual (30)

    - ✓ Technical Contribution: git logs & code quality (10)
    - ✓ Work Quality (10)
    - ✓ Presentation (10)
    - ✓ Peer Review ±5

PREPARED BY

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